

ORIGINAL ARTICLE

Adjuvant Cemiplimab or Placebo in High-Risk Cutaneous Squamous-Cell Carcinoma

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ABSTRACT

BACKGROUND

Patients who have cutaneous squamous-cell carcinoma with high-risk features are at risk for recurrence after definitive local therapy. The benefit of systemic adjuvant therapy options has not been well established in clinical trials.

METHODS

In a phase 3, randomized trial, we enrolled patients with local or regional cutaneous squamous-cell carcinoma, after surgical resection and postoperative radiotherapy, at high risk for recurrence owing to nodal features (extracapsular extension with largest node ≥ 20 mm in diameter or at least three involved nodes) or nonnodal features (in-transit metastases, T4 lesion [with bone invasion], perineural invasion, or locally recurrent tumor with ≥ 1 additional risk feature). Patients were assigned in a 1:1 ratio to receive adjuvant cemiplimab (350 mg) or placebo, administered intravenously every 3 weeks for 12 weeks, followed by a dose increase to 700 mg administered every 6 weeks for up to 36 weeks (≤ 48 weeks total). The primary end point was disease-free survival. Secondary end points included freedom from locoregional recurrence, freedom from distant recurrence, and safety.

RESULTS

A total of 415 patients were assigned to cemiplimab (209) or placebo (206). The median follow-up was 24 months. Cemiplimab was superior to placebo with respect to disease-free survival (24 vs. 65 events; hazard ratio for disease recurrence or death, 0.32; 95% confidence interval [CI], 0.20 to 0.51; $P < 0.001$). The estimated 24-month disease-free survival was 87.1% (95% CI, 80.3 to 91.6) with cemiplimab and 64.1% (95% CI, 55.9 to 71.1) with placebo. Cemiplimab led to lower risks of locoregional recurrence (9 events, vs. 40 with placebo; hazard ratio, 0.20; 9% CI, 0.09 to 0.40) and distant recurrence (10 vs. 26 events; hazard ratio, 0.35; 95% CI, 0.17 to 0.72). Adverse events of grade 3 or higher occurred in 23.9% of the patients who received cemiplimab and in 14.2% of those who received placebo; discontinuation due to adverse events occurred in 9.8% and 1.5%, respectively.

CONCLUSIONS

Adjuvant cemiplimab therapy led to longer disease-free survival than placebo among patients at high risk for recurrence of cutaneous squamous-cell carcinoma. (Funded by Regeneron Pharmaceuticals and Sanofi; C-POST ClinicalTrials.gov number, NCT03969004.)

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*A list of the investigators in the C-POST trial is provided in the Supplementary Appendix, available at NEJM.org.

This article was published on May 31, 2025, at NEJM.org.

DOI: 10.1056/NEJMoa2502449

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CUTANEOUS SQUAMOUS-CELL CARCINOMA is the second most common skin cancer, with an estimated annual incidence of 2.4 million cases worldwide.¹ Surgery with curative intent is the centerpiece of the clinical management of cutaneous squamous-cell carcinoma, with cure in approximately 95% of patients.²⁻⁵ However, a subset of patients with cutaneous squamous-cell carcinoma have disease recurrence, either locoregional or distant, after undergoing surgery and receiving adjuvant radiotherapy.⁶ A randomized, phase 3 trial (Postoperative Skin Trial/Trans-Tasman Radiation Oncology Group [POST/TROG] 05.01) showed no additional benefit of carboplatin administered concurrently with adjuvant radiotherapy, as compared with radiotherapy alone, in patients at elevated risk for recurrence of cutaneous squamous-cell carcinoma.⁶ However, that trial helped to identify patient subpopulations at the highest risk for recurrence.^{6,7}

Cemiplimab, a programmed death 1 (PD-1)–targeting antibody, is approved for the treatment of locally advanced (i.e., not suitable for resection) or metastatic cutaneous squamous-cell carcinoma, with a response occurring in 47% of patients and an estimated median duration of response of 41 months (range, 2 to 55).^{8,9} The C-POST trial is a phase 3, randomized trial comparing adjuvant cemiplimab with placebo in patients at high risk for recurrence of cutaneous squamous-cell carcinoma after surgery and postoperative radiotherapy. We report here the results of the primary analysis, which was conducted after more than approximately half the events expected for the final analysis of disease-free survival had occurred.

METHODS

PATIENTS

We recruited patients from 107 sites across 16 countries. Eligible patients were 18 years of age or older with local or regional cutaneous squamous-cell carcinoma and had completed both curative-intent surgery, with macroscopic gross resection of all disease, and postoperative radiotherapy (or concurrent chemoradiotherapy) at a biologically equivalent dose of at least 50 Gy within 2 to 10 weeks before randomization. Eligible patients had high-risk nodal or nonnodal features (or both) (Fig. S1 in the Supplementary

Appendix, available with the full text of this article at NEJM.org). High-risk nodal disease was defined as extracapsular extension with at least one node measuring at least 20 mm in diameter or as at least three involved nodes regardless of extracapsular extension. High-risk nonnodal disease was defined as any of the following: in-transit metastases, radiologic or clinical evidence of perineural invasion of named nerves, T4 primary tumor (with bone invasion), or local recurrence with at least one other adverse feature (nodal stage $\geq$ N2b, $\geq$ T3 lesion [diameter, $>$ 4.0 cm], or poorly differentiated histologic characteristics with recurrent lesion measuring $\geq$ 20 mm in diameter).

Patients were excluded if they had concurrent cancer (other than localized cutaneous squamous-cell carcinoma and certain low-risk diagnoses that were permitted according to the protocol), had received a solid-organ or stem-cell transplant previously, had clinically significant autoimmune disease, or had received any previous immunotherapy for cutaneous squamous-cell carcinoma. The full list of inclusion and exclusion criteria is provided in the protocol, available at NEJM.org.

TRIAL DESIGN AND TREATMENT

C-POST is an ongoing, international, randomized, phase 3 trial (Fig. S2). In part 1 of the trial (double-blind design; hypothesis-testing portion), patients were randomly assigned in a 1:1 ratio to receive either cemiplimab or placebo. In the original protocol, the regimen involved administration “every 3 weeks only,” with cemiplimab at a dose of 350 mg or placebo administered intravenously every 3 weeks. In protocol amendment 2 (June 29, 2021), the regimen was revised to “start at a frequency of every 3 weeks, with a switch to every 6 weeks.” Under this regimen, cemiplimab at a dose of 350 mg was administered intravenously every 3 weeks for 12 weeks, followed by cemiplimab at a dose of 700 mg administered intravenously every 6 weeks for an additional 36 weeks, or placebo was administered intravenously every 3 weeks for 12 weeks, followed by placebo administered intravenously every 6 weeks for an additional 36 weeks. The planned duration for the trial regimen was up to 48 weeks or until disease recurrence, the occurrence of unacceptable toxic effects, or withdrawal of consent, whichever was earlier. In part 2 of the trial (unblinded design), patients in the placebo group

who had disease recurrence or those in the cemiplimab group who had disease recurrence that occurred at least 3 months after the completion of part 1 had the option to receive subsequent cemiplimab.

Randomization was performed according to a central randomization scheme provided by an interactive Web-response system. Randomization was stratified according to tumor location (head and neck vs. non-head and neck), geographic region (North America vs. Australia or New Zealand vs. the rest of the world), high-risk category (nodal vs. nonnodal), Eastern Cooperative Oncology Group performance-status score (0 vs. 1; on a scale from 0 to 5, with higher scores indicating greater disability), and history of chronic lymphocytic leukemia (yes vs. no).

TRIAL OVERSIGHT

The trial was conducted in accordance with the principles of the Declaration of Helsinki, Good Clinical Practice guidelines of the International Council for Harmonisation, and all applicable regulatory requirements. The protocol was approved by the relevant institutional review boards or ethics committees at each site. All the patients provided written informed consent. An independent data monitoring committee evaluated data approximately every 6 months and provided oversight of the trial.

The trial was sponsored by Regeneron Pharmaceuticals and Sanofi and was designed by employees of Regeneron Pharmaceuticals in collaboration with the Trans-Tasman Radiation Oncology Group, with the first author as the lead investigator. Data were collected by the trial investigators, analyzed by statisticians employed by Regeneron Pharmaceuticals, and interpreted by the authors. Medical writing and editorial assistance with an earlier version of the manuscript was provided, in accordance with Good Publication Practice Guidelines, by a medical writer who was employed by Regeneron Pharmaceuticals. Sanofi approved the trial design but was not involved in data collection or analysis, the preparation of the manuscript, or the decision to submit the manuscript for publication. The authors were responsible for all content and editorial decisions. The authors vouch for the completeness and accuracy of the data and for the fidelity of the trial to the protocol.

END POINTS AND ASSESSMENTS

The primary end point was disease-free survival, defined as the time from randomization to the first documented disease recurrence (locoregional or distant) or death due to any cause. Secondary end points included freedom from locoregional recurrence, freedom from distant recurrence, overall survival, second primary cutaneous squamous-cell carcinoma tumors, and safety. Second primary tumors were defined as new cutaneous squamous-cell carcinoma lesions arising on the skin that could be managed by local therapy as part of routine clinical practice. Protocol-specified exploratory end points included patterns of recurrence, patient-reported outcomes, and correlations between efficacy and tumor expression of programmed death ligand 1 (PD-L1). The trial end points are described in the protocol.

Patient-reported outcomes were evaluated with multiple instruments, including the European Organization for Research and Treatment of Cancer Quality-of-Life Questionnaire–Core 30 (EORTC QLQ-C30), a 30-item questionnaire consisting of five functional scales, nine symptom scales or items, and a global health status–quality of life scale. Scales are linearly transformed into scores ranging from 0 to 100, with higher scores for the functional and general health status–quality of life scales indicating better functional status and quality of life, respectively, and lower scores for the symptom scales indicating lower severity of symptoms. A change of at least 10 points in any scale is considered to be clinically meaningful.¹⁰ Assessment of tumor PD-L1 expression ($\geq 1\%$ vs. $< 1\%$ of tumor cells expressing PD-L1) according to the tumor proportion score as determined by means of immunohistochemical testing was performed as previously described.¹¹ Additional details of the EORTC QLQ-C30 instrument and scoring and the PD-L1 expression analyses are provided in the Supplementary Methods section in the Supplementary Appendix.

During the treatment period, radiologic assessments were performed at screening and at the end of each 12-week cycle. In the follow-up period, clinical and radiologic assessments were performed every 4 months for the first 2 years and every 6 months thereafter.

Safety was monitored at each visit, with assessment of adverse events that occurred during the treatment period. Adverse events were graded

according to the Common Terminology Criteria for Adverse Events, version 5.0, of the National Cancer Institute. Immune-related adverse events were classified according to a list defined by Regeneron Pharmaceuticals. Adverse events of special interest were infusion-related reactions of grade 2 or higher and immune-related adverse events of grade 3 or higher. Causality was assessed by the investigator.

STATISTICAL ANALYSIS

We planned for a total enrollment of 412 patients. On the basis of literature review and analysis of the POST/TROG 05.01 trial, a 3-year disease-free survival of 55% in the placebo group, with a hazard ratio of 0.6 for the comparison of cemiplimab with placebo, was assumed (see the protocol). We calculated that 165 events of disease recurrence or death (disease-free survival analysis) with three interim analyses (at approximately 83, 107, and 132 events) would provide the trial with 90% power to detect a significant between-group difference in disease-free survival at a two-sided alpha of 0.05. Prespecified interim analyses used the Lan–DeMets O’Brien–Fleming spending function to control for the type I error. This report presents data from the first interim analysis of the fully enrolled trial (data-cutoff date, October 4, 2024). Because this analysis crossed the prespecified efficacy threshold for disease-free survival, it became the primary analysis.

Efficacy analyses were conducted according to the randomized assignment (intention-to-treat approach). For the primary efficacy analysis of disease-free survival, hypothesis testing between the two groups was performed with the use of a stratified log-rank test. Hypothesis testing was not performed for secondary end points. For time-to-event analyses, hazard ratios and 95% confidence intervals were estimated with the use of a stratified Cox regression model. The stratification factors for log-rank tests and Cox regression models were tumor location (head and neck vs. non-head and neck) and geographic region (North America vs. Australia or New Zealand vs. the rest of the world). Subgroup analyses of disease-free survival estimated the between-group treatment effect and nominal 95% confidence interval in prespecified subgroups. Safety analyses were based on whether the patient received cemiplimab or placebo and were conducted in all the patients who received any cemiplimab or placebo.

Prespecified analyses of patient-reported outcomes included descriptive analyses and overall changes from baseline across treatment cycles, which were analyzed with the use of a mixed-effects model for repeated measures. These analyses were conducted in patients who had a baseline score and at least one postbaseline score for the patient-reported outcome. All the data were analyzed with the use of SAS software, version 9.4 (SAS Institute).

RESULTS

PATIENTS

From June 2019 through August 2024, a total of 526 patients at high risk for recurrence of cutaneous squamous-cell carcinoma underwent screening, and 415 were randomly assigned to receive adjuvant therapy with cemiplimab (209 patients) or placebo (206) (Fig. S3). Among all the patients, the median age was 71 years (range, 33 to 95), 83.9% were men, and 82.7% had primary cutaneous squamous-cell carcinoma of the head and neck (Table 1 and Table S1). The high-risk nodal disease category included 242 patients (58.3%). The most common high-risk criterion was extracapsular extension in at least one node measuring at least 20 mm in diameter (in 48.4% of the patients). Overall, the demographic and disease characteristics of the patients at baseline were well balanced between the two trial groups and were generally representative of patients with high-risk cutaneous squamous-cell carcinoma (Table S2).

Among the 409 patients who underwent randomization and received at least one dose of cemiplimab or placebo, the median duration of exposure was 47.9 weeks (range, 3 to 52) in the cemiplimab group and 47.7 weeks (range, 3 to 51) in the placebo group. Overall, 44 of 205 patients receiving cemiplimab (21.5%) and 61 of 204 patients receiving placebo (29.9%) discontinued the regimen during part 1 of the trial. In the cemiplimab group, the most common reasons for discontinuation were adverse events (in 19 patients), disease relapse (in 10), and patient decision to withdraw (in 9); in the placebo group, the most common reasons were disease relapse (in 50) and patient decision to withdraw (in 5).

Overall, the median potential follow-up from randomization to the data-cutoff date was 24 months (range, 2 to 64). As of the data-cutoff date,

69 patients (16.9%; 36 in the cemiplimab group and 33 in the placebo group) were still receiving cemiplimab or placebo in part 1 of the trial.

EFFICACY

A significant improvement in disease-free survival was seen in the cemiplimab group as compared with the placebo group (24 vs. 65 events; hazard ratio for disease recurrence or death, 0.32; 95% confidence interval [CI], 0.20 to 0.51; $P < 0.001$) (Fig. 1 and Table S3). The Kaplan–Meier curves for the analysis of disease-free survival separated early and remained so for the duration of follow-up. The estimated disease-free survival at 24 months was 87.1% (95% CI, 80.3 to 91.6) in the cemiplimab group and 64.1% (95% CI, 55.9 to 71.1) in the placebo group.

With regard to the 65 events of disease recurrence or death in the placebo group, 61 patients had disease recurrence and 4 died without disease recurrence; with regard to the 24 events in the cemiplimab group, 18 patients had disease recurrence and 6 died without disease recurrence. The disease-free survival benefit with cemiplimab in relevant subgroups is shown in Figure 2.

Cemiplimab treatment prolonged freedom from both locoregional and distant recurrences as compared with placebo (Fig. 3 and Table S4). The estimated percentage of patients free from locoregional recurrence at 24 months was 94.6% (95% CI, 89.1 to 97.3) in the cemiplimab group and 76.7% (95% CI, 69.1 to 82.6) in the placebo group. Locoregional recurrence occurred in 9 patients in the cemiplimab group and in 40 in the placebo group (hazard ratio, 0.20; 95% CI, 0.09 to 0.40). The estimated percentage of patients free from distant recurrence at 24 months was 94.3% (95% CI, 89.0 to 97.1) in the cemiplimab group and 83.8% (95% CI, 76.3 to 89.0) in the placebo group. Distant recurrence occurred in 10 patients in the cemiplimab group and in 26 in the placebo group (hazard ratio, 0.35; 95% CI, 0.17 to 0.72). Patterns of disease recurrence are provided in Table S5.

Exploratory analyses of disease-free survival according to the two dose regimens that were used in this trial appeared to favor cemiplimab over placebo (Table S6). Additional exploratory analyses showed that the disease-free survival benefit with cemiplimab as compared with placebo appeared to be maintained regardless of tumoral PD-L1 status (among 85 patients with a PD-L1

tumor proportion score of $<1\%$: 8 vs. 16 events [hazard ratio, 0.32; 95% CI, 0.12 to 0.86]; among 309 patients with PD-L1 tumor proportion score of $\geq 1\%$: 14 vs. 45 events [hazard ratio, 0.28; 95% CI, 0.15 to 0.52]) (Fig. S4).

Among patients with recurrent disease, the most common subsequent intervention was cemiplimab (Table S7). Among 46 patients who had been originally randomly assigned to the placebo group and received cemiplimab for recurrent disease in part 2 of the trial, 20 (43%) had an objective radiographic response (Table S8).

A total of 25 deaths had occurred as of the data-cutoff date: 12 deaths (4 in the cemiplimab group and 8 in the placebo group) were due to disease progression and 13 deaths (8 in the cemiplimab group and 5 in the placebo group) were due to other causes (Table S9). Overall survival at 2 years was 94.8% (95% CI, 89.6 to 97.4) in the cemiplimab group and 92.3% (95% CI, 86.5 to 95.7) in the placebo group. The hazard ratio for death was 0.86 (95% CI, 0.39 to 1.90), with 51 patients in the placebo group receiving cemiplimab after recurrence (Fig. S5). At a subsequent data-cutoff date of April 7, 2025, there were 33 deaths (15 in the cemiplimab group and 18 in the placebo group), with a hazard ratio of 0.78 (95% CI, 0.39 to 1.56).

SAFETY

Adverse events due to any cause during the treatment period occurred in 91.2% of the patients who received cemiplimab and in 89.2% of those who received placebo (Table 2 and Table S10). The most common adverse events with cemiplimab as compared with placebo were fatigue (in 22.0% vs. 21.6% of the patients), pruritus (in 16.1% vs. 12.3%), rash (in 16.1% vs. 8.8%), and diarrhea (in 15.6% vs. 18.6%). Adverse events of grade 3 or higher that were due to any cause occurred in 23.9% of the patients who received cemiplimab and in 14.2% of those who received placebo. Adverse events of grade 3 or higher that were considered by the investigator to be related to treatment occurred in 9.8% of patients in the cemiplimab group (Table S11). Adverse events, regardless of attribution, that led to death occurred in two patients in each group (Table 2). One death due to myositis in a patient receiving cemiplimab was considered by the investigator to be related to treatment.

Cemiplimab or placebo was discontinued

Table 1. Demographic and Clinical Characteristics of the Patients at Baseline.*

Characteristic	Cemiplimab (N=209)	Placebo (N=206)
Age		
Median (range) — yr	71 (33–87)	70.5 (36–95)
≥65 yr — no. (%)	153 (73.2)	141 (68.4)
Male sex — no. (%)	174 (83.3)	174 (84.5)
Race — no. (%)†		
Asian	5 (2.4)	8 (3.9)
White	189 (90.4)	189 (91.7)
Other	1 (0.5)	1 (0.5)
Unknown or not reported	14 (6.7)	8 (3.9)
Geographic region — no. (%)		
North America	37 (17.7)	31 (15.0)
Australia or New Zealand	90 (43.1)	90 (43.7)
Rest of the world	82 (39.2)	85 (41.3)
ECOG performance-status score — no. (%)‡		
0	133 (63.6)	131 (63.6)
1	76 (36.4)	75 (36.4)
Anatomical region of resected high-risk tumor — no. (%)		
Head and neck	166 (79.4)	177 (85.9)
Non-head and neck	43 (20.6)	29 (14.1)
High-risk category — no. (%)§		
Nodal	125 (59.8)	117 (56.8)
Nonnodal	84 (40.2)	89 (43.2)
High-risk criteria — no. (%)¶		
Nodal disease with extracapsular extension and ≥1 lymph node ≥20 mm in diameter	105 (50.2)	96 (46.6)
Nodal disease with ≥3 nodes positive on surgical pathology report, regardless of extracapsular extension	33 (15.8)	37 (18.0)
In-transit metastases	20 (9.6)	21 (10.2)
T4 lesion	17 (8.1)	16 (7.8)
Perineural invasion	32 (15.3)	32 (15.5)
Recurrent high-risk cutaneous squamous-cell carcinoma with ≥1 additional feature	55 (26.3)	50 (24.3)
≥N2b disease associated with the recurrent lesion	17 (8.1)	13 (6.3)
≥T3 lesion	37 (17.7)	29 (14.1)
Poorly differentiated histologic findings and diam- eter of recurrent lesion ≥20 mm	16 (7.7)	13 (6.3)
PD-L1 tumor proportion score — no. (%)		
≥1%	155 (74.2)	154 (74.8)
<1%	42 (20.1)	43 (20.9)
Indeterminate	12 (5.7)	9 (4.4)

* Percentages may not total 100 because of rounding. PD-L1 denotes programmed death ligand 1.

† Race was reported by the patient.

‡ Eastern Cooperative Oncology Group (ECOG) performance-status scores are on a scale from 0 to 5, with higher scores indicating greater disability.

§ Patients with both nodal and nonnodal disease were classified as having nodal disease.

¶ The total for the high-risk criteria column is more than 100% because tumors could have more than one high-risk criterion. In the cemiplimab group, 13 patients had both nodal features (nodal disease with extracapsular extension and at least one lymph node measuring ≥20 mm in the greatest dimension, plus nodal disease with at least three positive nodes on the surgical pathology report, regardless of extracapsular extension) and 16 patients in the placebo group had both nodal features. A T4 lesion indicates disease with bone invasion, and a T3 lesion a tumor diameter of more than 4.0 cm. Perineural invasion was defined as clinical or radiologic involvement of named nerves.

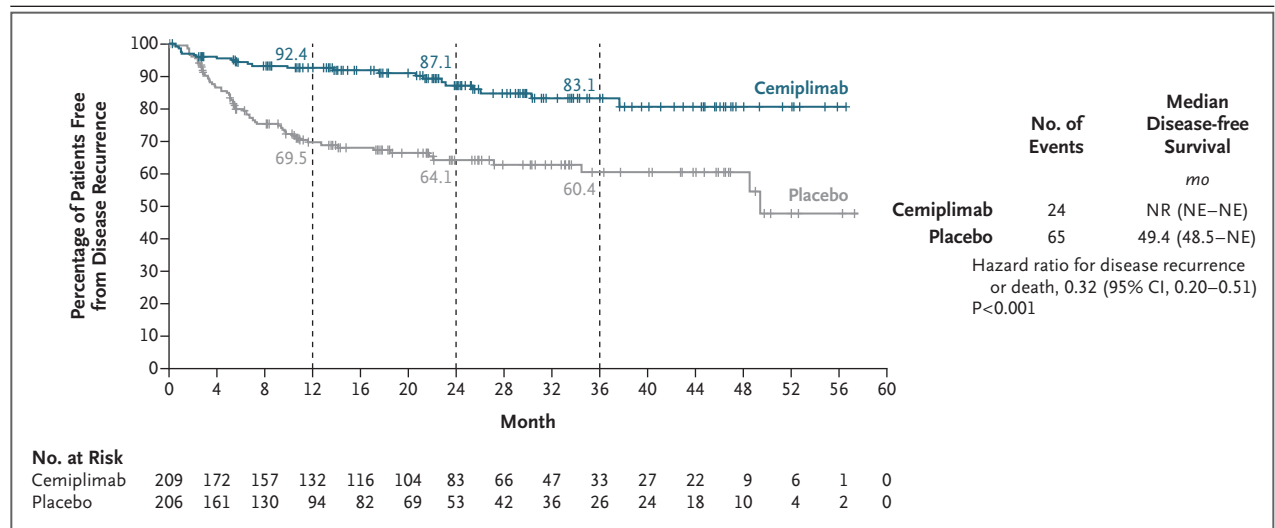


Figure 1. Disease-free Survival.

Analyses of disease-free survival were based on the Kaplan–Meier method, with stratification according to high-risk tumor (head and neck vs. non-head and neck) and geographic region (North America vs. Australia or New Zealand vs. the rest of the world). The threshold for significance was set to 0.00455 on the basis of the O’Brien–Fleming alpha spending function. The P value was based on a stratified proportional-hazards model. Second primary cutaneous squamous-cell carcinoma tumors were not included as events in the primary end-point analysis of disease-free survival. Tick marks indicate censored data. NE denotes could not be evaluated, and NR not reached.

owing to adverse events in 9.8% and 1.5% of the patients, respectively. Immune-related adverse events occurred in 22.9% of the patients who received cemiplimab (with events of grade ≥ 3 in 7.3%) and in 6.4% of those who received placebo (with no events of grade ≥ 3). No new immune-related adverse events were observed.

PATIENT-REPORTED OUTCOMES

More than 88% of the patients in each group completed the EORTC QLQ-C30 at baseline and through all the cycles. The overall change from baseline in EORTC QLQ-C30 global health status–quality of life scores across all time points during the treatment period indicated no meaningful between-group difference (difference in the least-squares mean change, -0.94 points [95% CI, -3.65 to 1.77]; clinically meaningful change, ≥ 10 points) (Table S12 and Fig. S6). Maintenance of the global health status–quality of life scores appeared to be sustained after the treatment period.

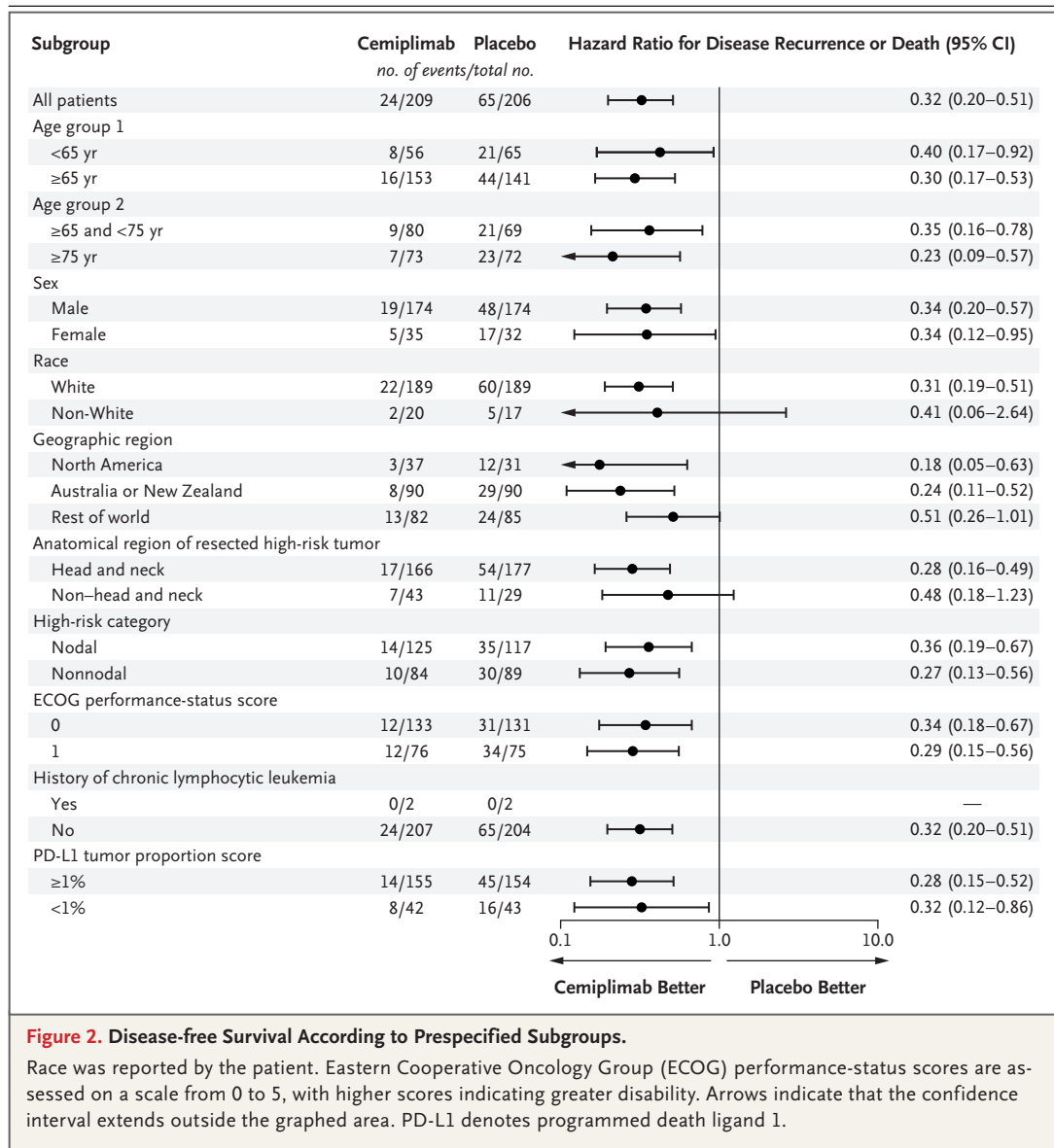
DISCUSSION

In this primary analysis of the phase 3 C-POST trial, disease-free survival was significantly longer with adjuvant cemiplimab therapy than with placebo among patients at high risk for recurrence

of cutaneous squamous-cell carcinoma after definitive local therapy. The risk of disease recurrence or death was 68% lower with cemiplimab than with placebo, with an estimated 24-month disease-free survival of 87% in the cemiplimab group and 64% in the placebo group.

Most recurrences were observed in the first year after surgical resection and the completion of adjuvant radiotherapy — findings that are consistent with the natural history of cutaneous squamous-cell carcinoma to recur rapidly.^{6,12} The Kaplan–Meier curves for disease-free survival diverged early, with continued separation throughout the follow-up period, thus indicating a rapid and sustained clinical benefit with cemiplimab.

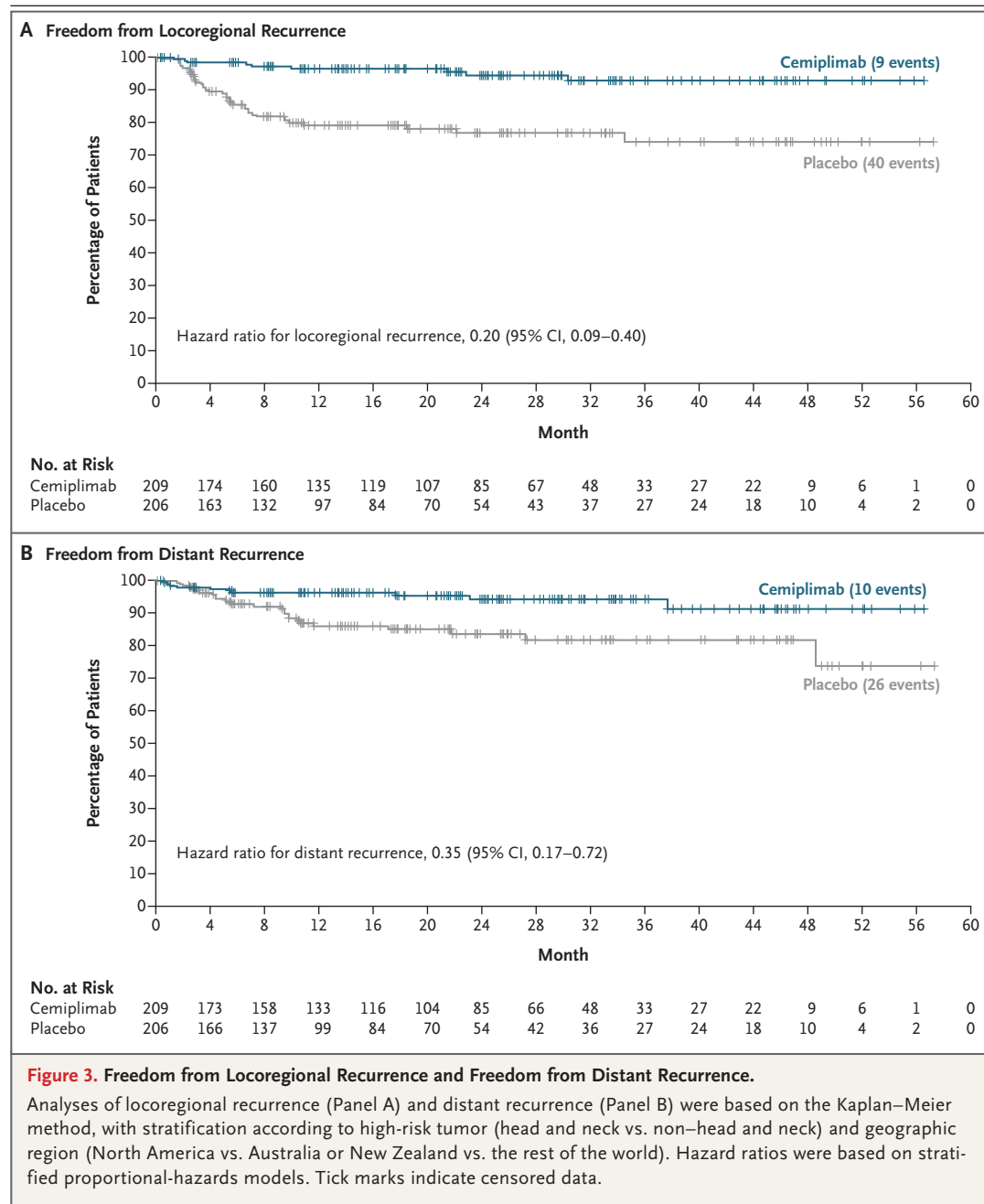
Analyses of patterns of recurrence showed that locoregional recurrences were more common than distant recurrences, a finding that was consistent with the results of the POST/TROG 05.01 trial.⁶ Locoregional recurrences are important medical events for patients with cutaneous squamous-cell carcinoma because they are associated with considerable risks of disease and death.^{13,14} In this trial, the risk of locoregional recurrence was 80% lower with adjuvant cemiplimab therapy than with placebo, and the risk of distant recurrence was 65% lower with cemiplimab than with placebo.



The safety profile of cemiplimab in the context of adjuvant therapy was consistent with the known safety profile of cemiplimab monotherapy in the context of advanced or metastatic disease. Discontinuation of the regimen due to adverse events occurred in 9.8% of the patients in the cemiplimab group. One death was considered by the investigator to be related to cemiplimab treatment. The global health status–quality of life score appeared to be maintained during treatment with adjuvant cemiplimab.

At the time of this primary analysis of disease-free survival, 25 deaths had been observed. A convincing benefit with regard to overall survival

has not been observed, although follow-up in this trial is ongoing. Most patients who had recurrent disease were treated subsequently with cemiplimab, which suggests that disease recurrences in this trial were usually considered by the treating physicians to be advanced cutaneous squamous-cell carcinoma. The availability of effective systemic therapy for recurrent disease may have an effect on the magnitude of any overall survival benefit from adjuvant therapy, as has been observed in melanoma.^{15,16} Individualized decision making is recommended with regard to whether to treat a patient with high-risk cutaneous squamous-cell carcinoma in the context of adju-



vant therapy or to wait until disease recurrence occurs before the initiation of immunotherapy; the circumstances and preferences of the patient should be taken into consideration. Because anti-PD-1 therapy provides durable responses in less than 50% of patients in the context of advanced cutaneous squamous-cell carcinoma,^{8,9,17,18} the ability of adjuvant cemiplimab to reduce the risk of recurrence of cutaneous squamous-cell carcinoma

is clinically meaningful for patients at high risk for recurrence.

A limitation of the C-POST trial is that it was not designed to formally investigate differences in efficacy and safety between the two dose regimens in the trial. However, this trial provides randomized data regarding the standard administration regimen (every 3 weeks only) and a regimen with an extended administration interval

Table 2. Adverse Events during Treatment Period, According to Grade.*

Event	Cemiplimab (N = 205)		Placebo (N = 204)	
	Any Grade	Grade ≥3	Any Grade	Grade ≥3
	<i>number of patients with event (percent)</i>			
Any adverse event	187 (91.2)	49 (23.9)	182 (89.2)	29 (14.2)
Serious adverse event	36 (17.6)	31 (15.1)	19 (9.3)	14 (6.9)
Adverse event leading to discontinuation of cemiplimab or placebo	20 (9.8)	16 (7.8)	3 (1.5)	2 (1.0)
Adverse event leading to death†	2 (1.0)	2 (1.0)	2 (1.0)	2 (1.0)
Adverse events in ≥10% of the patients in either group‡				
Fatigue	45 (22.0)	1 (0.5)	44 (21.6)	0
Pruritus	33 (16.1)	1 (0.5)	25 (12.3)	0
Rash	33 (16.1)	1 (0.5)	18 (8.8)	0
Diarrhea	32 (15.6)	3 (1.5)	38 (18.6)	0
Arthralgia	26 (12.7)	0	25 (12.3)	0
Hypothyroidism	24 (11.7)	1 (0.5)	6 (2.9)	0
Maculopapular rash	23 (11.2)	0	12 (5.9)	0
Bowen's disease	16 (7.8)	1 (0.5)	21 (10.3)	2 (1.0)

* Shown are adverse events that developed or worsened during the treatment period and any adverse events that were considered by the investigator to be related to cemiplimab or placebo that occurred during the posttreatment period but before part 2 of the trial (subsequent cemiplimab treatment).

† One death due to pneumonia was considered by the investigator to be unrelated to cemiplimab, and one death due to myositis was considered by the investigator to be related to cemiplimab. One death due to pneumonia and one death due to new primary malignant lung neoplasm were both considered by the investigator to be unrelated to placebo.

‡ Patients were counted only once according to the worst grade for multiple occurrences within a preferred term.

(start at every 3 weeks, with a switch to every 6 weeks). The trial results indicated that both regimens prolonged disease-free survival and had similar safety profiles.

Findings from the POST/TROG 05.01 trial⁶ were key determinants in defining the high-risk criteria and categories for the C-POST trial. The current trial was successful in defining a population of patients at high risk for recurrence of cutaneous squamous-cell carcinoma as evidenced by an estimated 3-year disease-free survival of approximately 60% in the placebo group, which closely matched the predicted disease-free survival that was based on data from the POST/TROG 05.01 trial. The results reported here will help to inform future updates to consensus staging criteria, as well as aid in the identification of patients in the clinic who are at the highest risk for disease recurrence.

Another phase 3 trial of adjuvant anti-PD-1 therapy in high-risk cutaneous squamous-cell

carcinoma was the KEYNOTE-630 trial of adjuvant pembrolizumab (ClinicalTrials.gov number, NCT03833167). That trial was stopped for futility after a prespecified analysis, according to a press release from the sponsor.¹⁹ Publication of the results of the KEYNOTE-630 trial is awaited and may provide information about why the trial did not show an improvement with pembrolizumab with regard to the primary end point.

In this trial, adjuvant cemiplimab therapy led to a large benefit, as compared with placebo, with regard to disease-free survival among patients at high risk for recurrence of cutaneous squamous-cell carcinoma.

Supported by Regeneron Pharmaceuticals and Sanofi. Dr. Rischin was supported in part by a National Health and Medical Research Council Investigator Grant (APP1175929).

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

We thank the trial participants, their families, and the trial-site staff; Anusha Dandu, M.S., Grace Fletchman, B.A., Fionnuala

Gallagher, M.Sc., Manish Harsiyani, B.Sc., Maria Rubino, B.S., and Fiby Thomas, M.D., of Regeneron Pharmaceuticals, for operational and data management assistance; Dimple A. Modi, Ph.D., for contributing to analyses of programmed death ligand 1 expression; Brian Head, Ph.D., of Regeneron Pharmaceuticals, for assistance with the development of the manuscript; and the staff of Alpha (a division of Prime, Knutsford, U.K.) for editorial assistance, funded by Regeneron Pharmaceuticals, with an earlier version of the manuscript.

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